SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 4

Name:
Student ID:

Due on Oct. 27, 2026, 18:00 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

A book has n typos. Two proofreaders, Prue and Frida, independently read the book. Prue catches each typo with probability p_1 and misses it with probability $q_1 = 1 - p_1$, independently, and likewise for Frida, who has probabilities p_2 of catching and $q_2 = 1 - p_2$ of missing each typo. Let X_1 be the number of typos caught by Prue, X_2 be the number caught by Frida, and X be the number caught by at least one of the two proofreaders.

(a) Find the distribution of X.

(b) For this part only, assume that $p_1 = p_2$. Find the conditional distribution of X_1 given that $X_1 + X_2 = t$.

Problem 2

Let X be a random day of the week, coded so that Monday is 1, Tuesday is 2, *etc.* (so X takes values $1, 2, \ldots, 7$ with equal probabilities). Let Y be the next day after X. Do X and Y have the same distribution? What is $P(X < Y)$?

Problem 3

For X and Y binary digits (0 or 1), let $X \oplus Y$ be 0 if $X = Y$ and 1 if $X \neq Y$ (this operation is called exclusive or (often abbreviated to XOR), or addition mod 2).

(a) Let $X \sim \text{Bern}(p)$ and $Y \sim \text{Bern}(1/2)$, independently. What is the distribution of $X \oplus Y$?

(b) With notation as in sub-problem (a), is $X \oplus Y$ independent of X? Is $X \oplus Y$ independent of Y? Be sure to consider both the case $p = 1/2$ and the case $p \neq 1/2$.

(c) Let $X_1, ..., X_n$ be i.i.d. (i.e., independent and identically distributed) Bern(1/2) R.V.s. For each nonempty subset J of $\{1, 2, ..., n\}$, let

$$Y_J = \oplus_{j \in J} X_j.$$

Show that $Y_J \sim \text{Bern}(1/2)$ and that these $2^n - 1$ R.V.s are pairwise independent, but not independent.

Problem 4

Let X be a discrete R.V. whose distinct possible values are $a_1, a_2, ..., a_n$, with probabilities $p_1, p_2..., p_n$, respectively (so $p_1 + p_2 + \cdots + p_n = 1$). The entropy of X is defined to be the average surprise of learning the value of X:

$$H(X) = \sum_{j=1}^{n} p_j \log_2(1/p_j).$$

Show that the maximum entropy for X is when its distribution is uniform over $a_1, a_2, ..., a_n$, i.e., $p_j = 1/n, \forall j$.